August 3, 2023

ASPEN AEROGELS

Q2 2023 FINANCIAL RESULTS CALL

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These risks and uncertainties include, but are not limited to, the following: inability to execute the growth plan through 2025, inability to manage supply chain disruptions to avoid undue delay or impact on operations or construction of Plant II and the Mexico assembly facility, inability to create new product, partnership and market opportunities; any sustained downturn in the industry or energy prices; any sustained downturn in the energy, energy infrastructure, chemical and refinery, LNG, sustainable building materials, EV thermal barrier, EV battery materials or other markets; any failure to sustain project-based demand in the subsea, LNG, on-shore or other markets; the right of EV thermal barrier customers to cancel contracts with Aspen at any time and without penalty; any costs, expenses, or investments incurred by Aspen in excess of projections used to develop pricing under the contracts with EV thermal barrier customers; any failure of Aspen or PyroThin to meet contractual specifications and requirements under contracts with EV thermal barrier customers; Aspen's inability to create customer or market opportunities for, including PyroThin; any other battery performance and safety products, battery materials or for other new products developed from Aspen's aerogel technology; any disruption or inability to achieve expected capacity levels in any of our three existing production lines in East Providence, RI or the Mexico assembly facility; any inability to establish or timely establish thermal barrier assembly operations in Mexico or any other location; the failure to receive all regulatory or other approvals required to operate, maintain or expand any of Aspen's facilities; any failure to achieve demand for Aspen's products; any failure to achieve expected price increases or average selling prices for Aspen's products; any significant increase in the cost of raw materials, utilities or any other manufacturing consumable; shortages of raw materials, utilities or any other manufacturing consumable; the failure to generate sufficient operating cash flow or to obtain significant additional capital to pursue Aspen's strategy; any failure to timely raise sufficient capital to fund various capital projects; the failure of Aspen's products to become widely adopted; the competition Aspen faces in its business; any failure to enforce any of Aspen's patents; any failure to protect or expand Aspen's Aerogel Technology Platform™; any future finding of invalidity of any of Aspen's patents in any jurisdiction; any failure to generate sufficient operating cash flow or to obtain sufficient additional capital to continue to pursue Aspen's new business, technology, patent enforcement, or patent defense strategy; any failure of Aspen's products to meet applicable specifications and other performance, safety, technical and delivery requirements; the general economic conditions and cyclical demands in the markets that Aspen serves; the economic, operational and political risks associated with sales and expansion of operations in foreign countries, including Mexico; the loss of any direct customer, including distributors, contractors and OEMs; compliance with health and safety laws and regulations; the maintenance and development of distribution channels; and the other risk factors discussed under the heading "Risk Factors" in our Annual Report on Form 10-K for the year ended December 31, 2022 and filed with the Securities and Exchange Commission ("SEC") on March 16, 2023, as well as any updates to those risk factors filed from time to time in our subsequent periodic and current reports filed with the SEC. 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Q2 2023 Developments

Driving on path to profitability by leveraging existing assets and optimizing costs while making commercial progress

Q2 2023 Highlights

- Delivered 17% gross margins
- 27% Energy Industrial gross margin, the highest in segment history
- Meaningful progress towards additional PyroThin® customers
- Well positioned to supply General Motors volume ramp in the second half of 2023

Near Term Potential Financial Profile

Leveraging existing assets and opportunities to deliver:

$550M	$200M	$140M
Annual Revenue Capacity	Annual Potential Gross Profit	Annual Potential Adjusted EBITDA

The Road to Creating Near Term Value

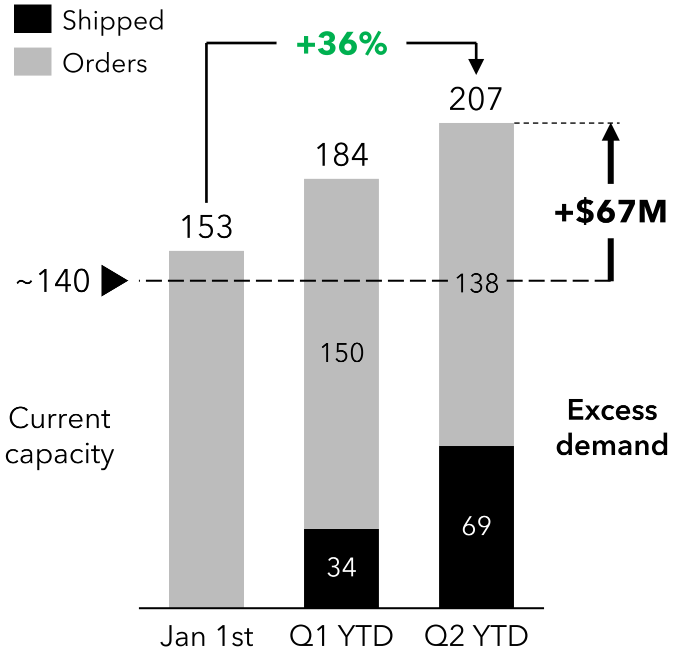
- Minimize and Right-time Near Term CAPEX
- Reduce Fixed Conversion Costs
- Secure Additional OEM Thermal Barrier Awards
- Optimize OPEX and Hold it Flat
- Deliver Expected EV Thermal Barrier Demand
- Start Energy Industrial Supplemental Supply

Energy Industrial Overview

Strong demand for Energy Industrial products, across all regions and applications, is currently supply constrained

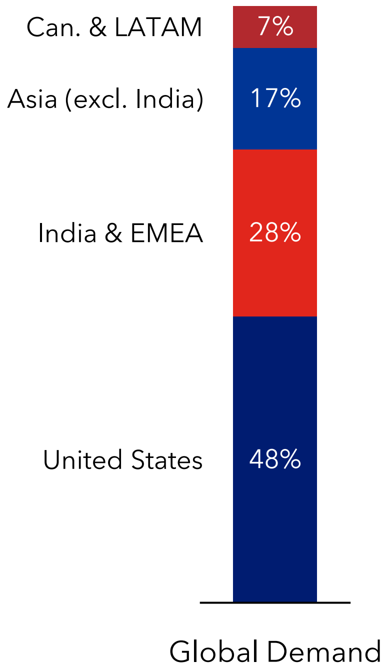
Energy Industrial demand is outpacing our current supply...

Energy Industrial FY 2023
Shipped + Open Purchase Orders ($M)



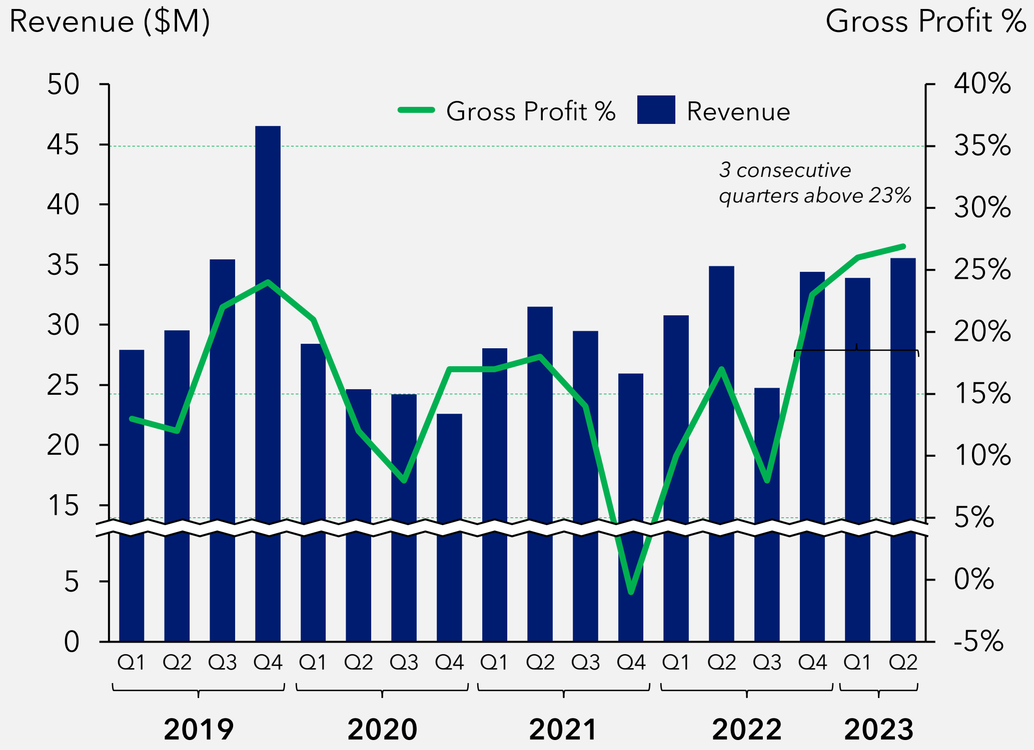
… driven by global demand…

Energy Industrial demand
% of 2023 Total Demand



…while Q2 2023 gross profit margin reaches all-time high as progress towards 35%+ target continues

Energy Industrial Historical Financials
2019 – Q2 2023 Revenue ($M) and Gross Profit % Trends



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Q2 and H1 2023 Financial Highlights

Material and fixed cost reductions enabled a 41% YoY reduction in our adjusted Q2 EBITDA loss on 6% higher revenues.

$ in millions except per share figures	Q2 2022	Q2 2023[1]	H1 2022	H1 2023[1]	Commentary
REVENUES YoY Growth	$45.6 44%	$48.2 6%	$84.0 41%	$93.7 12%	EV Thermal Barrier revenues in H1 of $24.3M reflect an expected delay in the ramp of GM's EV Thermal Barrier part demand. Energy Industrial revenues of $69.4M reflect supply constrained volume, updated pricing and a product mix aimed at throughput.
Net (Loss)	$(24.1)	$(15.4)	$(43.5)	$(32.2)	Operating efficiencies from prior investments and enhanced production capabilities drove positive gross margin, while cost discipline in OPEX will contribute to near-term profitability.
Adj. EBITDA[2]	$(18.3)	$(10.8)	$(33.0)	$(24.8)	Currently optimizing production plans for profitability by manufacturing products that deliver the highest revenue per standard hour of fixed manufacturing costs.
EPS	$(0.68)	$(0.22)	$(1.27)	$(0.47)	Continued investments to deliver on increasing customer demand.
Cap Ex	$37.9	$66.0	$52.4	$115.4	$40.7M spent towards Plant II in Q2; remainder invested in EV Thermal Barrier assembly tooling in Mexico, Rhode Island Aerogel plant improvements, and our labs in the Boston area.

1- Unaudited financials for 2023
2- See slide 12 herein for a reconciliation of net loss, the most directly comparable GAAP measure, to Adjusted EBITDA for the presented period

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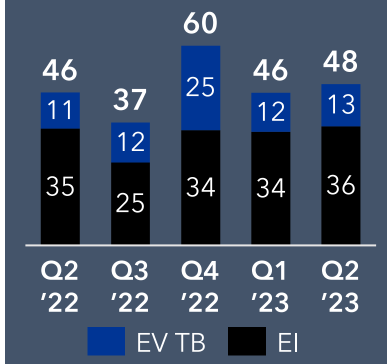
Historical Trends and 2023 Outlook

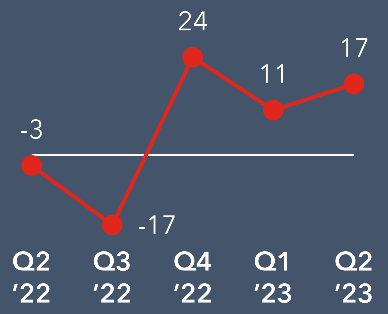
Improved profitability guidance, while wide revenue outlook range is driven by variability in Q4 EV Thermal Barrier demand

15 Month Quarterly Trajectory[1]

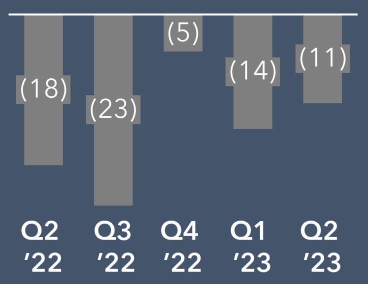
Revenues - $M



Gross Profit Margin %



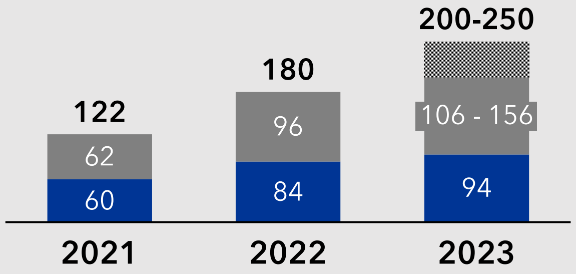
Adj. EBITDA[2] - $M



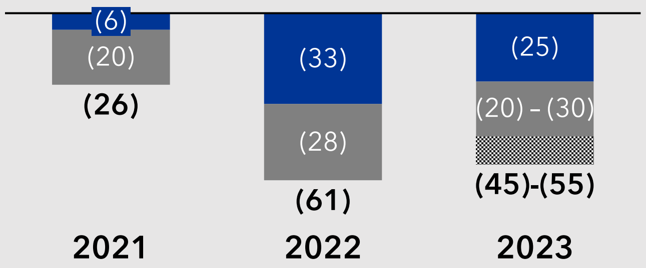
3 Year Trajectory[1]

1H 2H

Revenues - $M



Adj. EBITDA[2] - $M



2023 Outlook[1]

$ in millions except per share figures

REVENUES YoY Growth	$200.0 - $250.0 11% 39%
Net (Loss) *Prior Range*	$(85) - $(75) *$(102) - $(92)*
Adj. EBITDA[2] *Prior Range*	$(55) - $(45) *$(60) - $(50)*
EPS *Prior Range*	$(1.21) - $(1.07) *$(1.46) - $(1.31)*
Cap Ex	Managing to $150

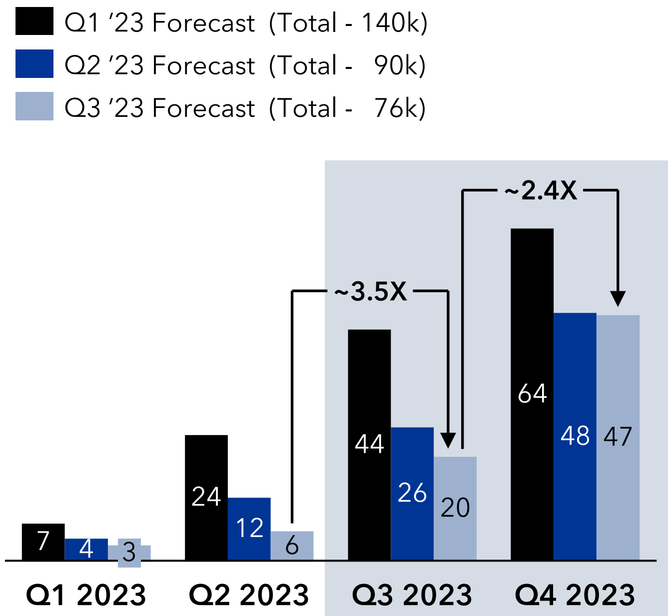
1- Unaudited financials for 2023
2- See slide 12 herein for a reconciliation of net loss, the most directly comparable GAAP measure, to Adjusted EBITDA for the presented outlook
Note: Figures may not total due to rounding

EV Thermal Barrier Demand Overview

GM Ultium production is expected by IHS' forecast to increase over 3X from Q2 to Q3, and 2.4X from Q3 to Q4

Ultium platform seeing short-term delays, volumes to accelerate in Q4

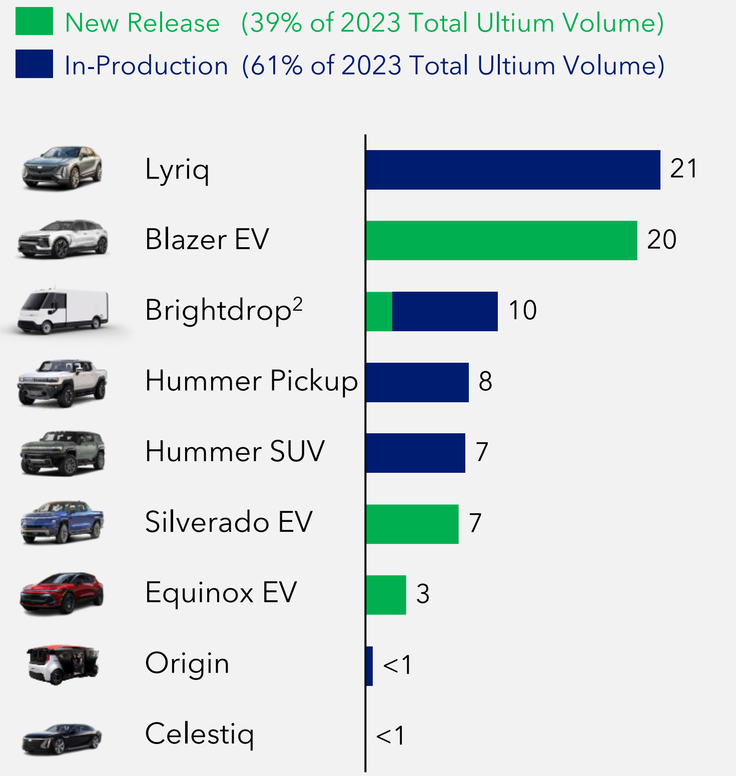
2023 GM Ultium EV Production Forecast[1]
(Vehicles in thousands)



89%
of the ~76k 2023 Ultium volumes in H2

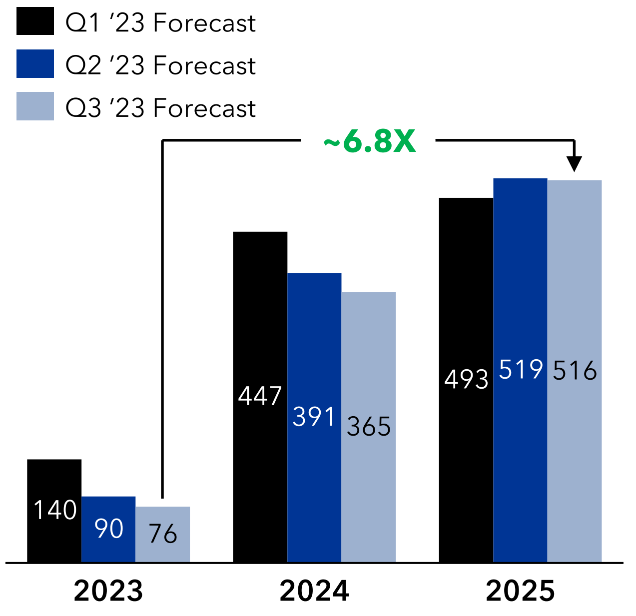
Major GM nameplates expected to launch, current nameplates to scale

2023 GM Ultium EV Production Forecast[1]
(Vehicles in thousands from Q3 '23 forecast)



While the long-term outlook remains strong

GM Ultium EV Production Forecast[1]
(Vehicles in thousands)



1- IHS January 2023, April 2023, July 2023 vehicle production forecasts
2. Brightdrop includes Zevo 400 (Q4 2023 launch) and Zevo 600 (in-production)

Near-Term Operating Plan Positioning

Focused on building a $550M+ revenue business, targeting 25% EBITDA margins with our current asset base

$ in millions, expressed as Annual Run Rate		Q2-22	Q3-22	Q4-22	Q1-23	Q2-23	LTM Q2-22	LTM Q2-23
EV Thermal Barrier Revenue Capacity	$400M	43	48	101	47	51	25	62
Energy Industrial Revenue Capacity	$150M	140	99	138	136	142	121	129
Total Annual Revenue Capacity	$550M	183	147	238	182	193	146	190
Material Costs[1]	40-45%	57%	57%	41%	41%	36%	53%	43%
Mfg. Costs[1]	20-25%	46%	61%	35%	48%	46%	48%	46%
Potential Gross Profit	35% 200	(3%)	(17%)	24%	11%	17%	(1%)	11%
OPEX	<20% Up to 110	86	88	96	96	102	67	95
Potential EBITDA[1]	25% 140	-73	-93	-18	-56	-43	-53	-53

Reflects supplemental GM order

Recent performance improvement

$30M of EBITDA loss reduction on $10M of incremental revenues

1. *Numbers are a percentage of total revenue*

Note: Figures may not total due to rounding

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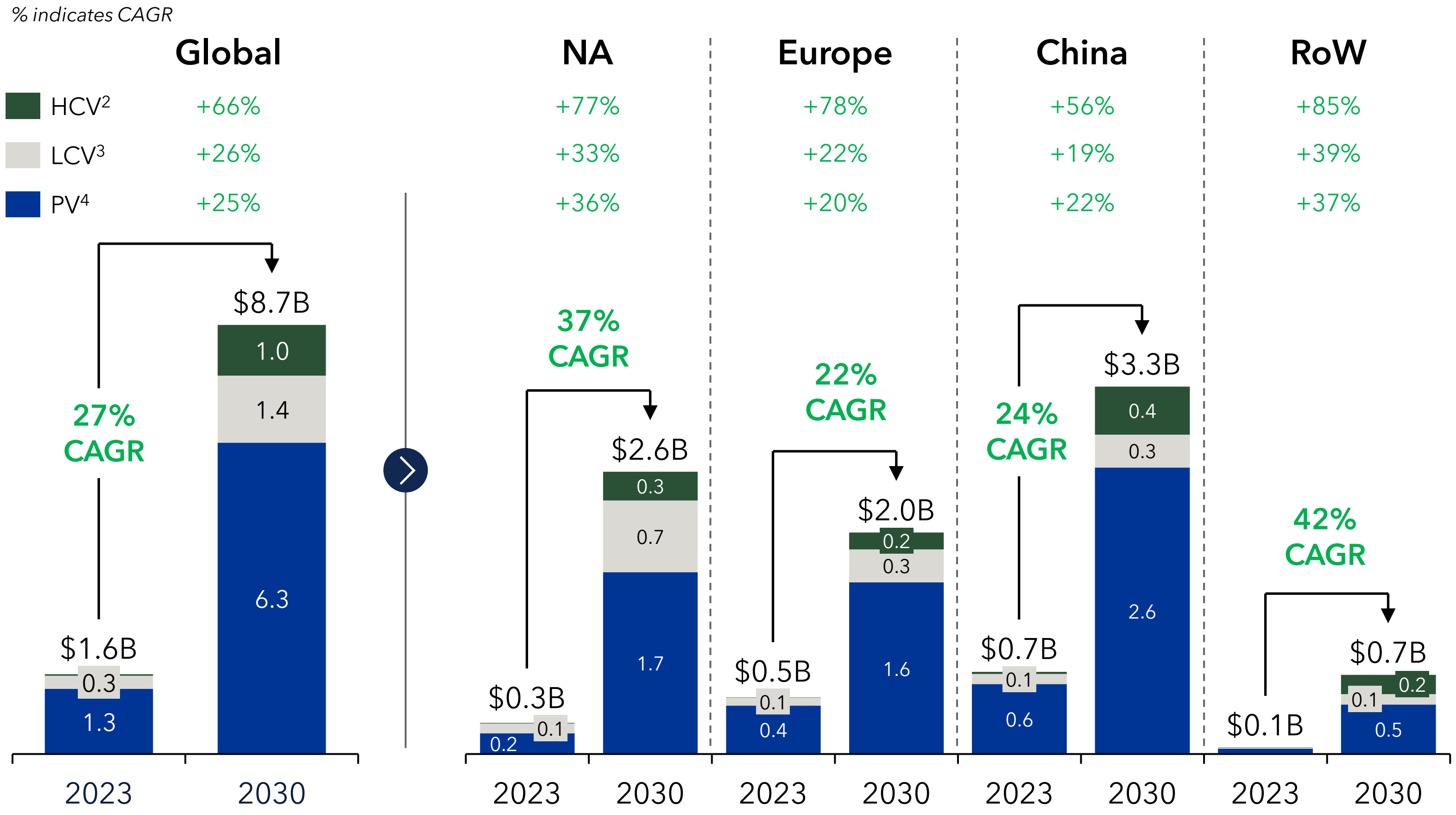
Building the EV Thermal Barrier Market

Serviceable addressable market of pouch and prismatic cell packs of $1.6B growing at 27% CAGR to $8.7B by 2030

Global serviceable addressable thermal barrier market[1] ($B USD)

Includes Cell-to-Cell Barriers for all BEV/PHEV Pouch and Prismatic battery cell designs across all battery chemistries



Key takeaways

North America
High growth rate attributed to larger battery pack size and heavily reliance on pouch cells; LCV is a critical segment of market in comparison to other regions

Europe
EV regulations result in higher base vs. NA; all OEMs with aggressive electrification plans, including large commercial vehicles

China
Largest serviceable market today, lower average CPV due to higher prismatic cell penetration

Rest of World
Small market (largely APAC region) with broad set of OEMs and battery strategies

Note: Figures may not total due to rounding 1. Includes cell-to-cell thermal barrier market for all pouch and prismatic vehicles including - passenger vehicles, light commercial vehicles, and heavy commercial vehicles 2. Heavy commercial vehicles, including buses, heavy-duty trucks, and medium-duty trucks 3. Light commercial vehicle 4. Passenger vehicle
Source: Aspen Aerogels analysis

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Summary

Earning our right to win by leveraging the Aerogel Technology Platform®, executing, and investing to grow profitably

ACCELERATING NEAR TERM PROFITABILITY

Leveraging existing assets and commercial opportunities to deliver $550M of annual revenue capacity, $200M of gross profit, and $140M of EBITDA

INCREASING FLEXIBILITY TO MEET DEMAND

Executing supplemental supply agreement to provide additional capacity for strong Energy Industrial demand, while existing assets fulfill EV OEMs

ADVANCED PYROTHIN® COMMERCIAL PROGRESS

GM volume ramp expected in 2H 2023, particularly in Q4; clear line of sight to additional PyroThin® awards from other OEMs

DEMONSTRATED OPERATING EXECUTION

Q2 gross profit highlights improving operating efficiencies, including record level Energy Industrial gross profit margins

Appendix

GAAP to Non-GAAP Reconciliation

	Q2		Q2 YTD		FY 2023 Outlook	
($ in thousands)	2023	2022	2023	2022	Low	High
Net Loss	(15,423)	(24,050)	(32,219)	(43,534)	(85,000)	(75,000)
Depreciation and Amortization	3,503	2,032	6,207	4,161	20,200	20,200
Stock-based Compensation	2,710	2,295	4,977	4,123	11,100	11,100
Interest (Income) Expense	(1,621)	1,404	(3,733)	2,264	(1,300)	(1,300)
Adjusted EBITDA	**(10,831)**	**(18,319)**	**(24,768)**	**(32,986)**	**(55,000)**	**(45,000)**

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PyroThin® Thermal Barrier Opportunity Drivers

Eight different macro parameters drive our content per vehicle (CPV) opportunity, and long-term revenue potential

#	Parameter	Options	Commentary
1	Vehicle sales by region	North America; Europe; China; Rest of World	• China expected to lead global EV penetration • Higher price and content vehicles in NA and EU
2	Vehicle classification	On-road: Light-duty (PV and LCV); Heavy Commercial Vehicles — Off-road	• Demonstrated commercial success within Light-duty • Most recent award in M/HDT segment highlights commercial vehicle opportunity potential
3	Powertrain technology	BEVs; PHEV; HEV; MHEV; FCEV; ICE	• TP risk increases as OEMs pack more energy density into EV batteries, both in EV's and Plug-In Hybrids • OEMs seeking to provide more range in PHEVs
4	Cell chemistry	LFP; LMFP; Ni-low/mid; Ni-rich; Semi solid-state	• All battery chemistries have thermal propagation risk • Highest CPV in nickel-based chemistries due to higher energy density
5	Cell form factor	Prismatic; Pouch; Cylindrical	• Pouch and Prismatic expected to continue to account for ~70% of global market • Cylindrical less suited for cell-to-cell solutions
6	Battery capacity	PC BEV 50 – 95 kWh; PC PHEV 15 – 30 kWh; LCV BEV 40 – 145 kWh; LCV PHEV 15 – 65 kWh; M/HDT BEV 150 - 400 kWh; M/HDT PHEV ~100 kWh	• OEMs seeking to increase range, while higher density packs have more risk of thermal propagation • NA and EU vehicles have larger batteries
7	Use case	Battery thermal barrier: Cell-to-cell; Module level; Pack level — Cabin insulation; Body noise dampening	• Aerogel solutions excel as a cell-to-cell barrier • Core requirements are thermal isolation, fire protection and mechanical properties
8	TR/TP mitigation goals	No TP Goals; Delay TP; Stop TP	• Recalls and global regulation expected to continue to driving OEM awareness and Stop TP goals • OEM goals to Stop TP results in more content

Core Focus Area — Potential Focus Area — Currently Out of Scope


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